

Forensic Bruise Segmentation: SegFormer vs YOLO with Knowledge Distillation

Technical

Report

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Abstract

This report presents a complete experimental pipeline for binary bruise segmentation in forensic imagery. We train five models – a SegFormer-B2 teacher, SegFormer-B0 (direct and distilled), and YOLO26n-sem (direct and distilled) – evaluated on a fixed held-out test set of 185 images. We address three research questions: (1) Can distillation transfer a large teacher’s accuracy to a smaller student? (2) Can YOLO’s near-binary probability maps be made amenable to threshold analysis via temperature scaling? (3) How do SegFormer and YOLO compare on speed under fair benchmarking conditions? SegFormer-B0 distilled achieves the best accuracy (Dice = 0.787), even surpassing the B2 teacher, while YOLO is approximately $2.4\times$ faster at 183–188 FPS.

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1 Introduction and Motivation

Bruise segmentation in forensic medical imaging is safety-critical: missing a bruise (false negative) can have serious clinical and legal consequences. Two competing architectures are evaluated:

- **SegFormer** – a transformer-based model pre-trained on ImageNet, producing soft probability maps amenable to threshold tuning, but computationally heavier.
- **YOLO26n-sem** – a single-stage CNN optimised for speed, pre-trained on Cityscapes, whose BCE-trained head produces near-binary probability maps requiring temperature scaling before threshold analysis is meaningful.

Knowledge distillation transfers the B2 teacher’s soft knowledge to both smaller student architectures.

2 Dataset and Experimental Setup

2.1 Dataset splits

All images were split into train, val, and a **fixed held-out test set** (185 images) before any training began. The test set was never used for training, hyperparameter search, or threshold selection – it was opened exactly once at the final evaluation stage.

Table 1: Dataset summary

Split	Images	Purpose
Train	–	Model training
Val	134	Threshold / alpha / hyperparameter search
Test	185	Final one-time evaluation

2.2 Common training configuration

Table 2: Shared hyperparameters

Parameter	Value
Image size	640×640
Epochs (max)	100
Early stopping patience	15
Optimizer (SegFormer)	AdamW
Backbone LR	6×10^{-5}
Head LR	6×10^{-4} ($10 \times$ backbone)
Weight decay	0.01
LR schedule	Linear warmup (1%) + polynomial decay
AMP	Enabled
Optimizer (YOLO)	Auto (Ultralytics)
YOLO LR final ratio	0.01 with cosine annealing

3 Model Architectures

3.1 SegFormer architecture

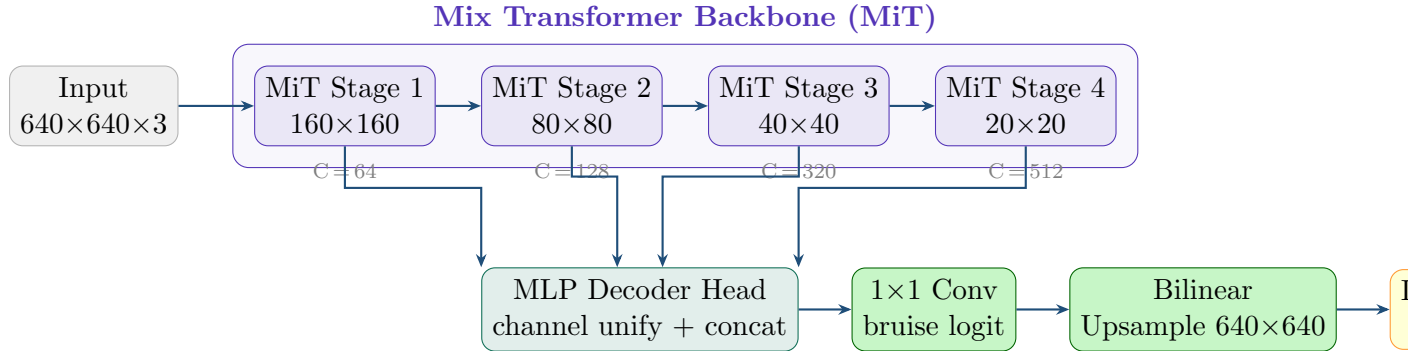


Figure 1: SegFormer architecture. Four hierarchical transformer stages produce multi-scale features unified by a lightweight MLP decoder head, followed by a 1×1 classifier and bilinear upsampling to the input resolution. All SegFormer models share this structure; B2 has larger channels and more attention heads than B0.

3.2 YOLO26n-sem architecture

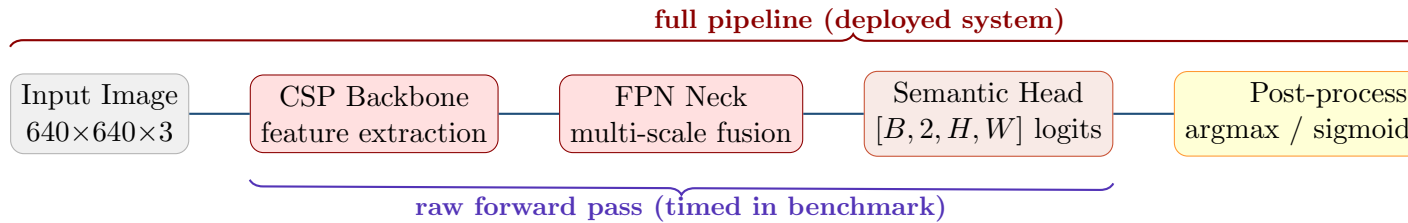


Figure 2: YOLO26n-sem architecture. Two timing scopes are shown: the raw forward pass covers only the neural network computation (fair for architecture comparisons); the full pipeline includes all pre- and post-processing.

3.3 Model parameter counts and pre-training

Table 3: Model summary

Model	Parameters	Pre-trained on	Role
SegFormer-B2	27.3 M	ImageNet	Teacher
SegFormer-B0	3.7 M	ImageNet	Student (direct & distilled)
YOLO26n-sem	–	Cityscapes	Student (direct & distilled)

4 Training Pipeline

4.1 Full five-model pipeline

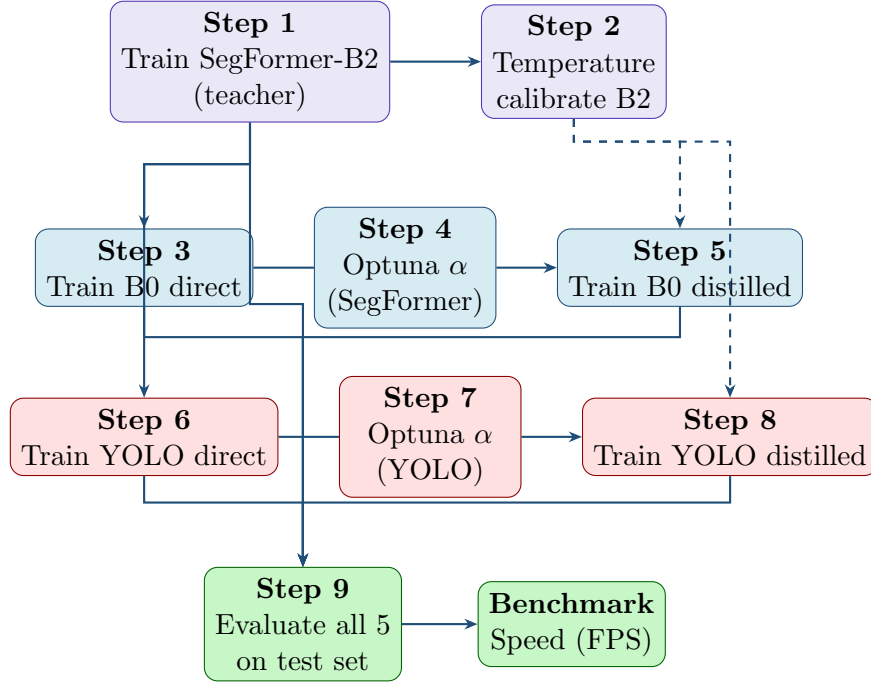


Figure 3: Full training pipeline. The B2 teacher is trained first and temperature-calibrated; its soft outputs guide distillation of both students. Optuna finds the optimal blending coefficient α for each student before the full-length distillation run.

4.2 Knowledge distillation

Distillation blends hard ground-truth labels with the teacher’s soft probability map using mixing coefficient α :

$$\tilde{y} = \alpha \cdot y_{\text{GT}} + (1 - \alpha) \cdot p_{\text{teacher}} \quad (1)$$

where $p_{\text{teacher}} = \sigma(z_{\text{teacher}}/T)$ is the temperature-scaled teacher probability.

Table 4: Distillation configuration

Model	Distillation method	Best α
SegFormer-B0 distilled	Soft loss (online, per step)	0.60
YOLO distilled	Hard pseudo-mask (offline, pre-computed)	0.40

4.3 Data preprocessing and augmentation

SegFormer pipeline

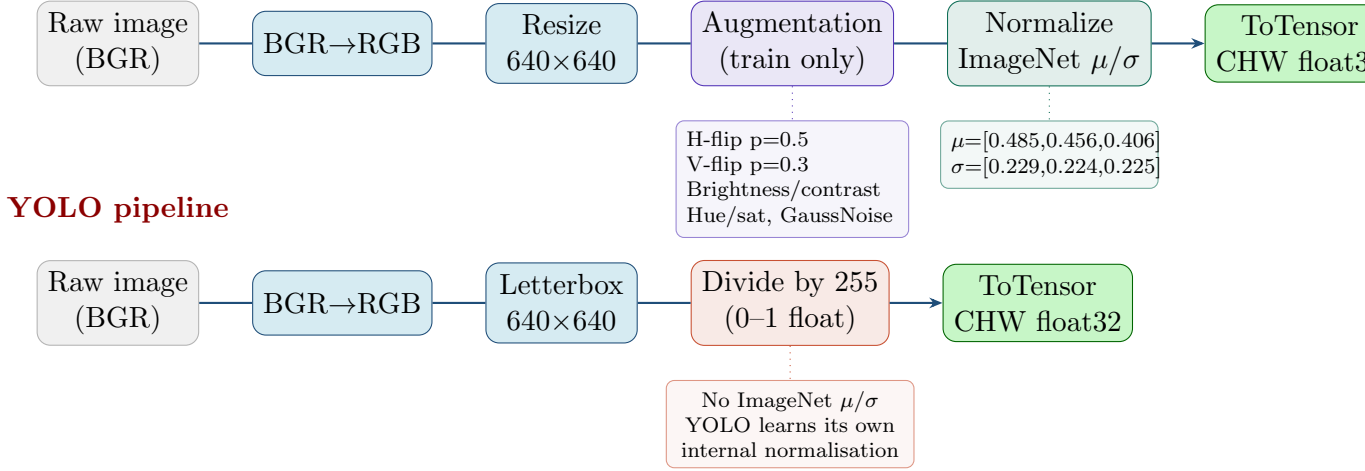


Figure 4: Preprocessing pipelines for SegFormer (top) and YOLO (bottom). SegFormer requires ImageNet mean/std normalisation because its backbone was pre-trained on ImageNet. YOLO only divides by 255; its backbone learns its own internal normalisation during training.

5 Threshold Selection and Temperature Scaling

5.1 SegFormer: threshold sweep on val

After training, SegFormer’s best checkpoint is evaluated on validation at 19 threshold values $\tau \in \{0.05, 0.10, \dots, 0.95\}$. The threshold maximising mean Dice is stored in `threshold_search.csv` and applied once on the test set.

5.2 YOLO: the near-binary problem

BCE loss is minimised when $\sigma(z) \rightarrow y \in \{0, 1\}$, requiring $|z| \rightarrow \infty$. The gradient update rule:

$$z \leftarrow z - \alpha \frac{\partial \mathcal{L}}{\partial z} = z - \alpha (\sigma(z) - y) \quad (2)$$

pushes logits toward $\pm\infty$ every step. After 100 epochs, logits reach $|z| \approx 5-8$, where sigmoid is essentially flat ($\sigma(7) = 0.999$). The probability histogram has two spikes near 0 and 1 with an empty gap – any threshold in that gap produces an identical binary mask, making threshold sweeping meaningless.

5.3 Temperature scaling as a de-saturation tool

Dividing logits by $T > 1$ before sigmoid pulls saturated values back into sigmoid’s non-flat region:

$$p_T = \sigma\left(\frac{z}{T}\right) \quad (3)$$

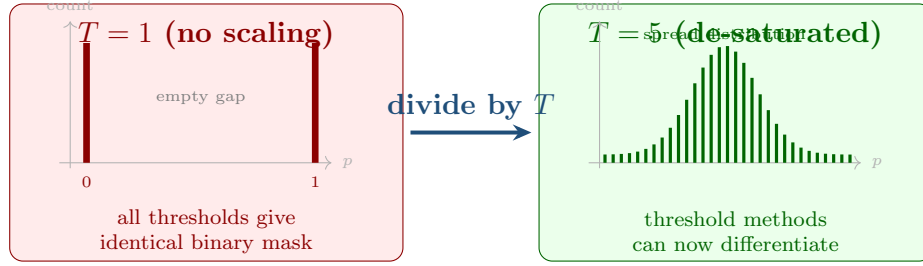


Figure 5: Effect of temperature scaling on YOLO’s probability histogram. At $T = 1$, probability mass clusters near 0 and 1; at $T = 5$, the distribution spreads, allowing threshold methods to differentiate.

5.4 Joint temperature–threshold sweep

For each YOLO model a 2D grid search is performed on val: $T \in \{1, 2, 3, 4, 5, 6, 7, 8\}$ crossed with $\tau \in \{0.05, \dots, 0.95\}$. The pair (T^*, τ^*) maximising val mean Dice is applied once on the test set.

Table 5: Val temperature–threshold search results

Model	T^*	τ^*	Val Dice	Note
YOLO direct	2.0	0.10	0.5645	$T = 1$ never in top results
YOLO distilled	6.0	0.40	0.6253	$T = 1$ nearly ties winner

6 Inference and Speed Benchmarking

6.1 Two FPS measurement contexts

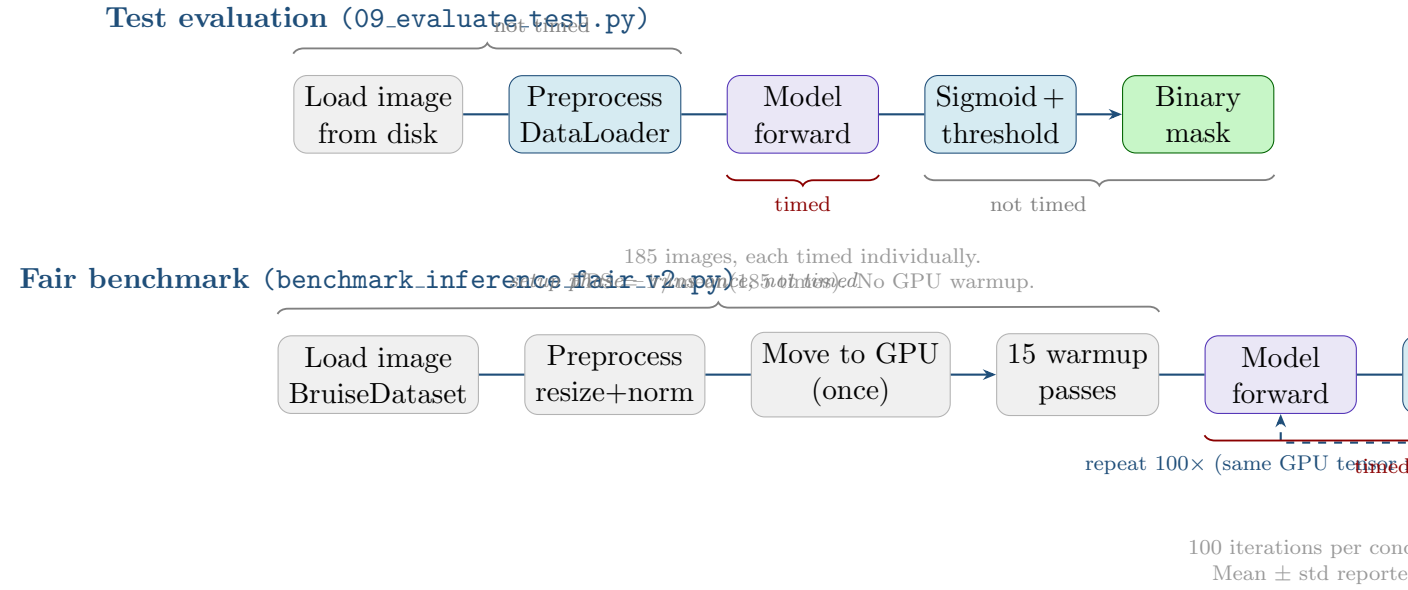


Figure 6: Two FPS measurement methodologies. Test-evaluation FPS reflects real-world per-image latency. Benchmark FPS is a controlled measurement under identical conditions for both models, with proper GPU warmup and the same preprocessed tensor reused across all timed iterations.

6.2 Raw forward vs. full pipeline

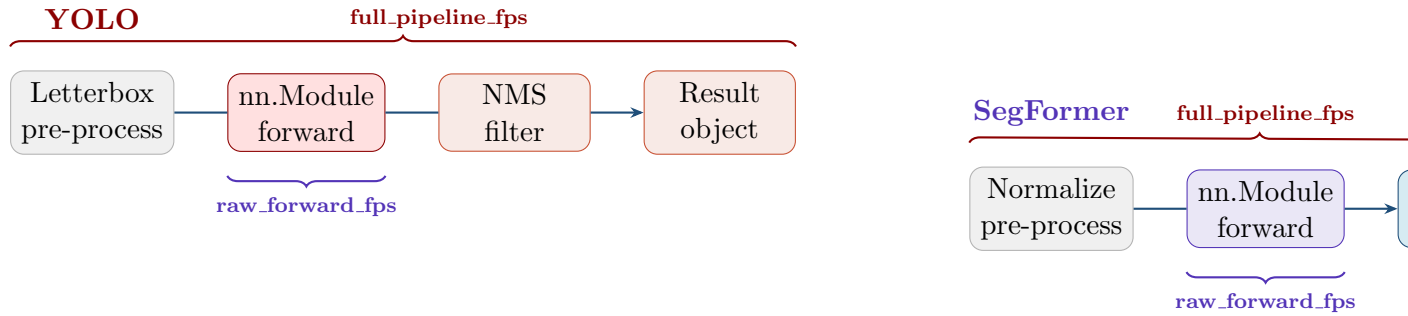


Figure 7: What is timed in each FPS category. The *raw forward* measures only the neural network computation – fair for architecture comparison. The *full pipeline* includes pre- and post-processing – the realistic deployed speed. For SegFormer, raw $\approx$ full (sigmoid is negligible). For YOLO, NMS can add 30–50% overhead.

6.3 Benchmark improvements over original

Table 6: Changes from original benchmark to v2

Issue	Original	v2 (fixed)
GPU warmup	None	15 warmup passes
Setup vs timing	Interleaved	Fully separated
SegFormer out-put	Raw logits only	Sigmoid + best_thr $\rightarrow$ binary mask
Image loading	Bespoke cv2 loader	BruiseDataset (same as training)
Statistics	Single mean	Mean $\pm$ std (100 iterations)
YOLO full pipeline	NMS only	NMS + yolo_sem_pred_mask()

7 Results

7.1 Accuracy on fixed test set (185 images)

Table 7: Test set accuracy – all five models

Model	Dice	IoU	Precision	Recall	Miss rate
SegFormer-B2 teacher	0.767	0.652	0.875	0.705	0.000
SegFormer-B0 direct	0.775	0.654	0.855	0.741	0.005
SegFormer-B0 distilled	0.787	0.672	0.853	0.763	0.005
YOLO direct	0.506	0.411	0.733	0.537	0.195
YOLO distilled	0.608	0.499	0.726	0.654	0.097

7.2 Speed benchmark (fp32, full pipeline, 640 px)

Table 8: Benchmark FPS – 100 iterations, GPU warmed up, mean $\pm$ std

Comparison	SegFormer FPS	YOLO FPS	Ratio
Direct (fp32)	77.8 $\pm$ 0.30 ms	187.9 $\pm$ 0.10 ms	2.41 $\times$
Distilled (fp32)	77.9 $\pm$ 0.22 ms	182.6 $\pm$ 0.10 ms	2.34 $\times$
Direct (fp16)	76.8 $\pm$ 0.18 ms	187.8 $\pm$ 0.09 ms	2.44 $\times$
Distilled (fp16)	76.3 $\pm$ 0.32 ms	186.8 $\pm$ 0.13 ms	2.45 $\times$

7.3 Accuracy vs. speed trade-off

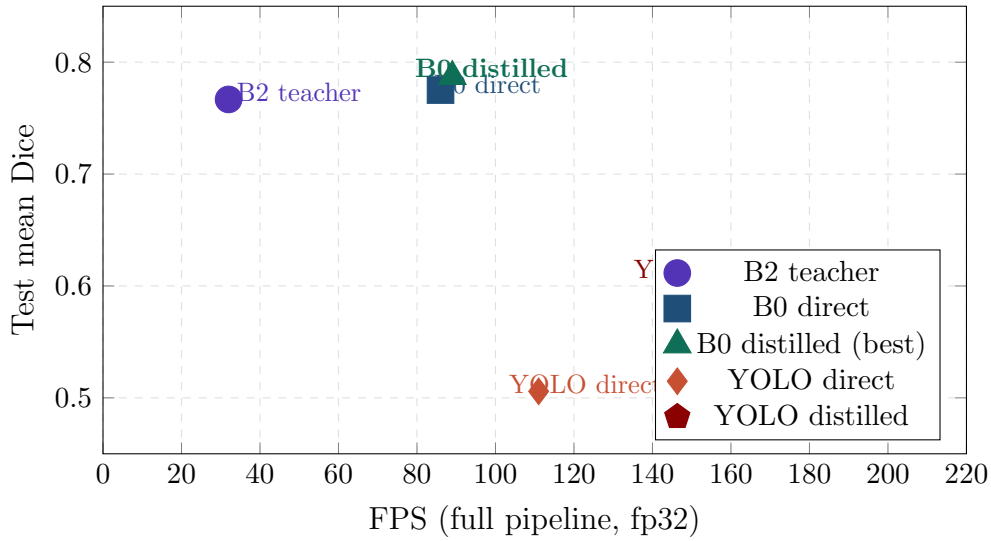


Figure 8: Accuracy vs. speed Pareto plot. SegFormer-B0 distilled (teal triangle) achieves the best Dice at a reasonable speed. B0 distilled even surpasses the B2 teacher, confirming successful knowledge transfer.

7.4 Impact of distillation

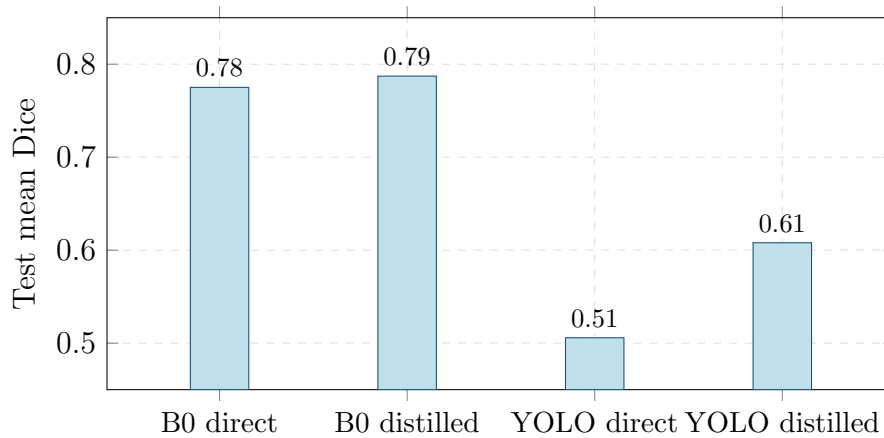


Figure 9: Effect of distillation on test Dice. SegFormer-B0 gains +0.012 and YOLO gains +0.102. YOLO benefits more proportionally because its direct training starts from a semantically distant domain (Cityscapes), and the B2 teacher’s soft pseudo-labels provide strong boundary supervision.

8 Key Findings and Discussion

8.1 Distillation outperforms the teacher

SegFormer-B0 distilled (Dice = 0.787) slightly outperforms the B2 teacher (0.767) despite having $7.3\times$ fewer parameters, consistent with the *born-again networks* phenomenon.

8.2 YOLO precision vs. recall trade-off

YOLO’s precision (≈ 0.73) is high for both variants but recall is significantly lower ($0.537 / 0.654$). In a forensic context where under-detection is the more dangerous failure mode, this is a significant limitation: YOLO direct misses bruise completely in nearly 1 in 5 images.

8.3 Temperature scaling confirms BCE saturation

YOLO direct required $T^* = 2.0$ to de-saturate probability maps; YOLO distilled was nearly insensitive to temperature ($T = 1.0$ almost tied $T^* = 6.0$). This directly confirms that distillation’s soft pseudo-labels prevent BCE-driven logit saturation.

8.4 fp16 provides no meaningful speedup

Neither model benefits from fp16 at batch size 1 on this GPU, suggesting the bottleneck is memory bandwidth rather than compute.

8.5 Speed-accuracy recommendation

Clinical recommendation

For a forensic bruise detection tool where missing a bruise has serious consequences, **SegFormer-B0 distilled** is the recommended deployment model. It achieves the best Dice (0.787), highest recall (0.763), a complete-miss rate of only 0.5%, and runs at 89 FPS – well above real-time. The $2.4\times$ speed advantage of YOLO is irrelevant when both models run in real time, and the accuracy gap (0.787 vs 0.608) is clinically significant.

9 Conclusion

We presented a complete five-model bruise segmentation pipeline covering training, knowledge distillation, temperature-based threshold calibration, and fair speed benchmarking. Key contributions:

1. **Distillation surpasses the teacher:** SegFormer-B0 distilled (Dice = 0.787) exceeds the B2 teacher (0.767) with $7.3\times$ fewer parameters.
2. **Temperature scaling enables YOLO threshold analysis:** BCE-trained models produce near-binary maps that make threshold sweeping meaningless without de-saturation; $T = 2$ suffices for the direct model.
3. **Distillation prevents logit saturation:** the distilled YOLO model needed almost no temperature correction, showing that soft pseudo-labels act as implicit calibration regularisation.
4. **Fair benchmarking matters:** the corrected benchmark (v2) with GPU warmup, BruiseDataset image loading, and mask-producing full pipeline timing gives substantially more trustworthy FPS numbers than the original implementation.